

# TCP Congestion Control: Principles, Phases, and Mathematical Dynamics

## Summary

TCP congestion control is designed to prevent **network-wide collapse** due to overwhelming retransmissions. Its core principle is to **regulate the sender's transmission rate** based on both **receiver capacity** and **network congestion**.

Key insights:

- Sender window is constrained by  $W_S = \min(W_R, W_C)$ , where  $W_R$  is the receiver's advertised window, and  $W_C$  is the congestion window.
- TCP adapts dynamically using three phases: **Slow Start**, **Congestion Avoidance**, and **Congestion Detection**.
- Congestion detection mechanisms include **Timeout Timers** (severe congestion) and **Three Duplicate ACKs** (mild congestion).
- Exponential growth in Slow Start transitions to linear growth in Congestion Avoidance, with thresholds controlling phase changes.

## 1. TCP Window Management Fundamentals

### 1.1 Flow Control (Receiver Limitation)

- **Receiver Window ( $W_R$ ):** Indicates the receiver's available buffer space (16-bit field in TCP header).
- Example: Receiver buffer = 1000 bytes, received 200 bytes →  $W_R$  advertised = 800 bytes.
- **Sender Rule:** Cannot exceed  $W_R$ ; ensures the receiver is not overwhelmed.

### 1.2 Congestion Control (Network Limitation)

- Before 1980s, TCP only considered  $W_R$ .
- **Network Congestion:** Occurs when routers cannot buffer incoming packets.
  - Dropped packets → no ACK → sender retransmits → traffic increases → potential congestive collapse.

### 1.3 Effective Sender Window

[  
W<sub>S</sub> = min(W<sub>R</sub>, W<sub>C</sub>)  
]

- **W<sub>R</sub>**: Receiver's advertised window.
- **W<sub>C</sub>**: Sender's estimated congestion window (network-aware).

## 2. Three Phases of Congestion Control

### 2.1 Slow Start Phase

**Objective:** Quickly increase transmission rate from a low initial value.

- **Initial W<sub>C</sub>**: 1 MSS.
- **Growth Pattern:**
  - Per RTT: doubles (exponential).
  - Per ACK: ( W<sub>C</sub> = W<sub>C</sub> + 1 ).
- **Transition:** Continues until **W<sub>C</sub> ≥ Ssthresh** (Slow Start Threshold).

**Example:**

|   | RTT | W <sub>C</sub> (MSS) |
|---|-----|----------------------|
| 1 |     | 1                    |
| 2 |     | 2                    |
| 3 |     | 4                    |
| 4 |     | 8                    |
| 5 |     | 16                   |

(Assume Ssthresh = 19 → next step jumps to 19 and transitions to linear growth)

### 2.2 Congestion Avoidance Phase

**Objective:** Increase throughput safely without overshooting network capacity.

- **Growth Pattern:**
  - Per RTT: (  $W_C = W_C + 1$  ) (additive increase).
  - Per ACK: (  $W_C = W_C + \frac{1}{W_C}$  ) (linear growth at byte-level granularity).
- **Constraint:**  $W_C \leq W_R$  (cannot exceed receiver buffer).

## 2.3 Congestion Detection Phase

**Mechanisms:**

| Feature                     | Timeout Timer<br>(Severe) | Three Duplicate ACKs (Mild)            |
|-----------------------------|---------------------------|----------------------------------------|
| Indicator                   | Timer expires before ACK  | 3 identical ACKs received              |
| Meaning                     | Severe congestion         | Mild congestion                        |
| New Threshold<br>(SSTHRESH) | $W_C / 2$                 | $W_C / 2$                              |
| New $W_C$                   | Reset to 1 (Slow Start)   | Set to SSTHRESH (Congestion Avoidance) |
| Special Action              | Standard retransmission   | Fast Retransmit (immediate)            |

**Fast Recovery:** For mild congestion, TCP avoids full reset and remains in linear growth.

## 3. Threshold and Transition Logic

- **SSTHRESH** is updated dynamically: half of the  $W_C$  at the point of congestion.
- **Transition from Slow Start to Congestion Avoidance:**
  - If doubling  $W_C$  exceeds SSTHRESH,  $W_C$  jumps to SSTHRESH and continues linear increase.

**Example:**

- Current  $W_C = 16$  MSS, SSTHRESH = 19 MSS

- Next Slow Start increment → 32 MSS exceeds threshold → W\_C set to 19 MSS → linear growth begins (20, 21, ...).

## 4. Maximum Window Constraints

- W\_C cannot exceed W\_R.
- Sender respects both **receiver buffer** and **network capacity**.
- Ensures that TCP remains **flow- and congestion-aware**, avoiding overloading the path.

## 5. Practical Implications

1. **Time Out → Slow Start**: Resets window to 1 MSS; network highly congested.
2. **Three Duplicate ACKs → Fast Recovery**: Sets W\_C to Ssthresh; network still partially functional.
3. TCP's dynamic adjustment prevents the **congestive collapse** observed in early networks.
4. Modern TCP variants (e.g., Reno, NewReno, Cubic) build upon this foundation with improved RTT estimation and congestion window growth strategies.